AMD DEVELOPER HACKATHON TRACK 3 UNICORN

Crossfire

CUDA to ROCm Translation Agent

An autonomous AI agent that ports CUDA code to AMD ROCm — HIPIFY-first migration, compiled and test-verified on AMD MI300X.

AMD CLOUD FIREWORKS AI GEMMA 4 12B ROCM 7.2

THE PROBLEM

AMD has the hardware. NVIDIA has the software moat. We breach it.

Multi-billion-dollar

AI INFRA MARKET LOCKED TO NVIDIA

Tens of thousands of ML teams want to migrate to AMD but cannot afford the rewrite.

2-8 wks

MANUAL CUDA-TO-ROCM PORT TIME

Per project. Every PyTorch extension, every custom kernel, every inference pipeline.

WHY NOW

Four forces converge in 2026

1. ROCm 7.2 is stable

ROCm has matured rapidly for production AI workloads. MIOpen, hipBLAS, rocPRIM are production-ready.

2. MI300X offers compelling inference economics

192GB HBM3, lower cost per token. The hardware advantage is real — the software gap is the only blocker.

3. Code LLMs are viable

Gemma 4, DeepSeek-Coder, and other modern code LLMs perform semantic code translation with RAG augmentation.

4. AMD says so themselves

Lisa Su: AMD continues to prioritize AI software and developer tooling. ROCm roadmap lists "CUDA porting tools" as priority.

THE SOLUTION

Autonomous translation + validation

Crossfire is an AI agent that takes a CUDA source files, translates it to ROCm, compiles it on AMD MI300X, runs it, and validates test verification against analytical baselines — autonomously, in minutes not weeks.

THE AGENT LOOP

Analyze Static analysis	→	**Translate** Hybrid routing	→	**Compile** hipcc on MI300X	→	**Run** AMD GPU	→	**Diff** vs. baseline	↻	**Done** validated

ARCHITECTURE

Every component is AMD-native

AMD

Agent Orchestrator

LangGraph state machine on AMD Developer Cloud. 6-node loop with retry.

AMD

Local Model

Gemma 4 12B via vLLM on MI300X. Local tokens = FREE. Prompt-engineering baseline; LoRA pipeline implemented.

AMD

Cloud Fallback

Gemma 4 12B via vLLM on MI300X for hard semantic cases (cuDNN, Triton, warp shuffle).

GEMMA

RAG Layer

ChromaDB with ROCm 7.2 docs, MIOpen API refs, translation tables.

AMD

Compile Sandbox

Docker + ROCm 7.2.3 + hipcc. Dockerized ROCm compile environment.

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Validation

Run on MI300X, numerical diff vs. CUDA baseline. Threshold: 1e-5.

LIVE DEMO

Watch it work

5

HERO DEMOS

From vector_add to warp-shuffle reduction

Each demo shows a distinct CUDA pattern: simple kernel, shared memory tiling, cuBLAS library call, PyTorch extension, warp shuffle primitives.

01	02	03	04	05
vector_add	matrix_mul	sgemm (cuBLAS)	pytorch_ext	warp_shuffle
Easy · 0.30	Medium · 0.45	Hard · 0.65	Hard · 0.70	Hardest · 0.90

→ Live demo at: `129.212.185.42:8000/ui`

VALIDATION

Validated on AMD MI300X

Every translation is compiled with hipcc, run on AMD MI300X, and numerically diffed against analytical baseline outputs. Pass threshold: max_abs_error < 1e-5.

1e-5	5	80%	$0
DIFF THRESHOLD	MAX ITERATIONS	LOCAL MODEL USAGE	LOCAL TOKEN COST

SAMPLE VALIDATION OUTPUT

```
$ crossfire translate vector_add.cu
✓ Analyzed: 1 kernel, difficulty=0.30, routing=LOCAL
✓ Translated: 1728 chars, 0 tokens (local, FREE)
✓ Compiled: hipcc success in 1.2s
✓ Ran on MI300X: 0.45ms
✓ Diff: max_abs_error=0.0e+00 → PASS
✓ Done in 2.1s, 1 iteration
```

TECHNICAL DIFFERENTIATION

HIPIFY-first + Gemma 4 12B semantic repair (LoRA pipeline on roadmap)

Approach	Syntax	Semantics	Validated	Time	AMD-Native
Manual rewrite	✓	✓	✓	2-8 wks	Optional
AMD HIPIFY (rule-based)	✓	✗	✗	Hours	✓
GitHub Copilot (generic)	Partial	Partial	✗	Hours	✗
Crossfire v1 prototype	✓	✓	✓	**Minutes**	✓

→ Crossfire combines semantic AI translation with autonomous validation on AMD hardware.

MARKET POTENTIAL

$50B market. A strategic bridge into the AMD ecosystem.

The Buyer

ML Platform Engineers at mid-to-large enterprises evaluating AMD hardware. Thousands of them. All blocked by CUDA lock-in.

The Revenue Model

Free tier (5 files/mo), Pro ($99/mo), Enterprise (custom). Plus AMD white-label / OEM licensing.

Strategic Fit

AMD has acquired Mipsology, Nod.ai, Silo AI — all software gap-closers. Crossfire fits the pattern exactly.

The Moat

Validated translation corpus grows with every job. Fine-tuned model improves. Each migration adds reusable repair traces and validated mappings.

EXECUTION

Built in 5 days. Imagine 5 months.

D1	D2	D3	D4	D5	D6
Kickoff	Agent core	Sandbox	Frontend	Polish	Submit
First translation	RAG + routing	ROCm compile + run	Demo + video	Edge cases	v1.0.0

Roadmap

v1.0 (Hackathon)

5 hero demos, single-file, prompt-engineered baseline

v2.0 (3 months)

Multi-file repos, fine-tuned model, 50K training pairs

v3.0 (12 months)

Triton support, cuDNN full coverage, enterprise SaaS

Crossfire is AMD's **strategic unlock.**

We built it in 5 days. Imagine what we build in 5 months with AMD.

AMD CLOUD · FIREWORKS · GEMMA 4 12B · ROCM 7.2

github.com/VampFay/CROSSFIRE_AMD-DEV-HACKATHON-ACT_2 · 129.212.185.42:8000/ui